

ANGELA KANG

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EDUCATION

University of Southern California

Applied Data Science, Masters of Science

Los Angeles, CA

August 2025 – May 2027(*expected*)

- **Relevant Courses:** Machine Learning for Data Science, Foundations of Data Management

University of California, Santa Barbara (UCSB)

Goleta, CA

Statistics and Data Science, Bachelors of Science

September 2021 – June 2025

- **Relevant Courses:** Probability & Statistics, Machine Learning, Multivariate Analysis, Time Series, Stochastic Processes, Linear Algebra, Real Analysis, Numerical Methods, Data Structures & Algorithms, Experimental Design, Python Programming

EXPERIENCE & LEADERSHIP

Hackathon Winner – SBHacks XI

- Collaborated with a 4-member team to design My Meowcho, a Chrome extension gamifying productivity with animated cats, customizable accessories, and sound effects.
- Led front-end development using JavaScript, HTML/CSS, and Chrome APIs
- Project won Best Overall Hack out of 60+ teams.

PROJECTS

Scout – AI Freelancer-Client Communication Platform | React, React Native, FastAPI, Python, WebSocket, RAG

- Built web frontend (React + Vite) and Android app (React Native) as Product Lead, delivering real-time WebSocket chat UI, project dashboard, and conversational onboarding flows across both platforms
- Engineered AI Scope Guardian UI (ScopeFlagCard, ChangeOrderPreview) surfacing RAG-based contract violation alerts and auto-generating change orders and invoices in real time
- Led end-to-end UX design and high-fidelity mockups for all major screens; implemented embeddings-based freelancer-client matching scored across skill fit, communication style, timeline, and budget alignment

COVID Custom SQL Engine | Python, Streamlit

- Designed and implemented core relational database operations (filter, select, groupby, join) from scratch using hash-based algorithms, without relying on pandas or SQL engines.
- Built a DataFrame-like abstraction to process and query large COVID-19 CSV datasets efficiently.
- Developed an interactive Streamlit dashboard to visualize time-series trends and support exploratory data analysis.

Jute Pest Classification | Python, TensorFlow, Transfer Learning

- Developed a deep learning image classifier for 17 jute pest classes using transfer learning on pre-trained CNN architectures.
- Compared ResNet50, EfficientNet, VGG16, and DenseNet201 using data augmentation and comprehensive evaluation metrics.
- Analyzed model performance to identify the most effective architecture for agricultural pest detection.

CUDA Matrix Convolution | CUDA, Python

- Implemented CPU and GPU versions of matrix multiplication and image convolution using CUDA for high-performance computing.
- Optimized memory access patterns and parallel execution to achieve significant speedups over CPU baselines.
- Integrated Python bindings to enable GPU-accelerated numerical operations within data science workflows.

APS Failure Prediction | Python, scikit-learn, XGBoost, SMOTE

- Built a predictive maintenance model for truck Air Pressure System failures on highly imbalanced data (1.67% failure rate).
- Applied SMOTE for class balancing and trained ensemble models including Random Forest and XGBoost.
- Achieved 99.31% accuracy and 0.9946 AUC while reducing false negatives critical to failure detection.

CERTIFICATIONS

Google Data Analytics Professional Certificate – Coursera, Authorized by Google (2025)

Google Business Intelligence Professional Certificate – Coursera, Authorized by Google (2025)

Google Data Analytics Professional Certificate – Coursera, Authorized by Google (2025)

SKILLS & INTERESTS

Languages: Python, R, SQL, C, CUDA, JavaScript, HTML, MATLAB, LaTeX, C++, React

Data Science & Analytics: Predictive modeling, statistical modeling, regression analysis, time series analysis, experimental design, ANOVA, machine learning, deep learning, transfer learning, exploratory data analysis (EDA), feature engineering, dimensionality reduction (PCA), imbalanced-data methods (SMOTE), clustering, relational data modeling, custom SQL-style operations (filter, select, groupby, join)

Tools & Frameworks: Pandas, NumPy, Matplotlib, Scikit-learn, TensorFlow, Git/GitHub, Jupyter, Excel, Streamlit, MySQL, MongoDB, Apache Spark, Hadoop, cuBLAS, BeautifulSoup, Selenium, Tesseract OCR, GCC, NVCC

Technical Skills: GPU programming & parallel computing, CUDA kernel development, high-performance computing, Linux/Ubuntu development, web scraping, data pipeline development, database querying, SQL joins and aggregations, NoSQL-style data handling, distributed data processing concepts, Python bindings to C/CUDA code, statistical analysis, data visualization